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# Aircraft conceptual design computational tool for static aeroelastic analysis

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# Introduction



## Static Aeroelasticity

- Studies the interaction between the aerodynamics and flexibility of the airframe
- Static aeroelastic phenomena: Divergence, Control Reversal, Elastic Loads ,etc.
- Impacts performance and airworthiness directly.
- Extremely relevant today: HALE UAV's, Composite Airframe, Sustainability and Efficiency
- Evident need for aeroelastic analysis in conceptual, and preliminary studies







# Introduction



## Literature Review

- **Optimization – Bowman (1989)**
  - 79% Mass reduction ⬇
  - 80% Lift and control efficiency increase ⬆
- **Composite Layup – Dilinger (2019)**
  - Analysis of different aerodynamic models in optimization
- **Low-Cost Analysis - Rajpal (2019)**
  - 89% Mass reduction in 42 iterations ⬇
- **Strong x Weak Coupling - Sholten and Hartl (2021)**
  - Strong – Direct integration of aerodynamic and structural model; Accuracy ⬆
  - Weak – Models solved in parallel; Computational Cost ⬇

## Objective

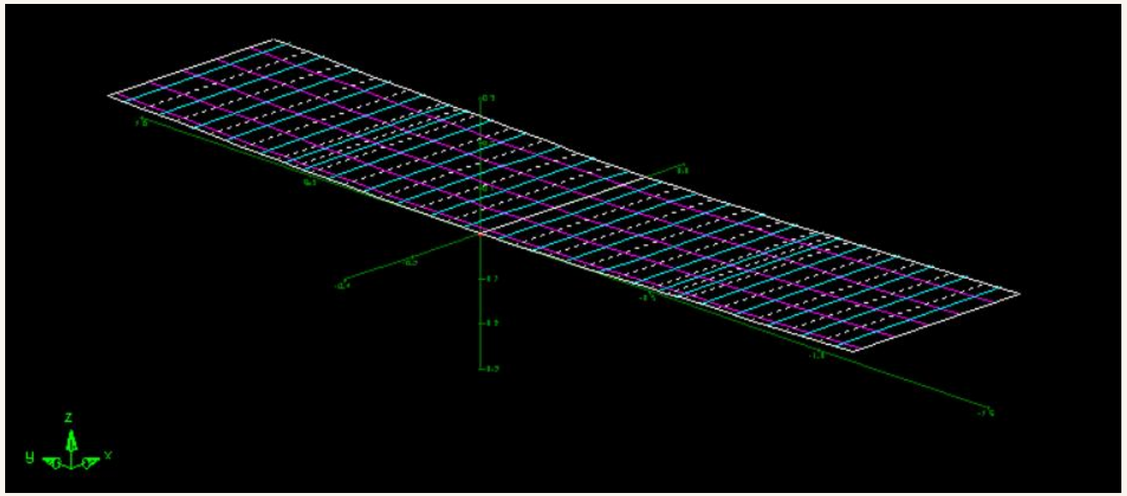
- **Development of a weakly coupled static aeroelastic solver of low computational cost**

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# Methodology

## Aerodynamic Model

- Vortex Lattice Method (VLM) in *Athena Vortex Lattice* (AVL)
  - Potential Flow
  - $[AIC]\{\Gamma\} = \{\overrightarrow{U_0}.\hat{\eta}\}$
- Implementation using *avlwrapper*
- Local $cl(y), cm(y), cd(y)$

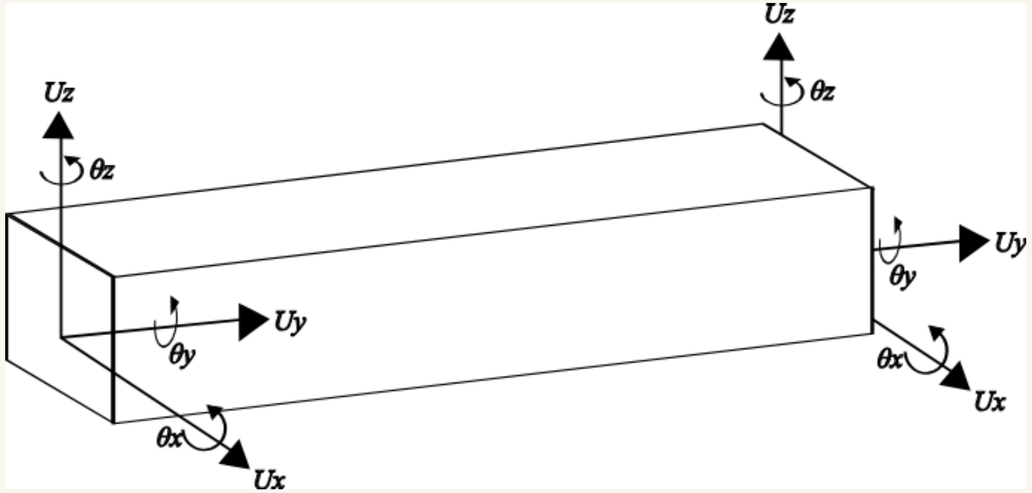


# Methodology



## Structural Model

- Finite Element Model – 12 *DOF* Beam Element (Displacements and Rotations)
- $[K]\{U\} = \{F(\Gamma)\}$
- Constant load distribution on each element
- Nodal loads obtained through work equivalence

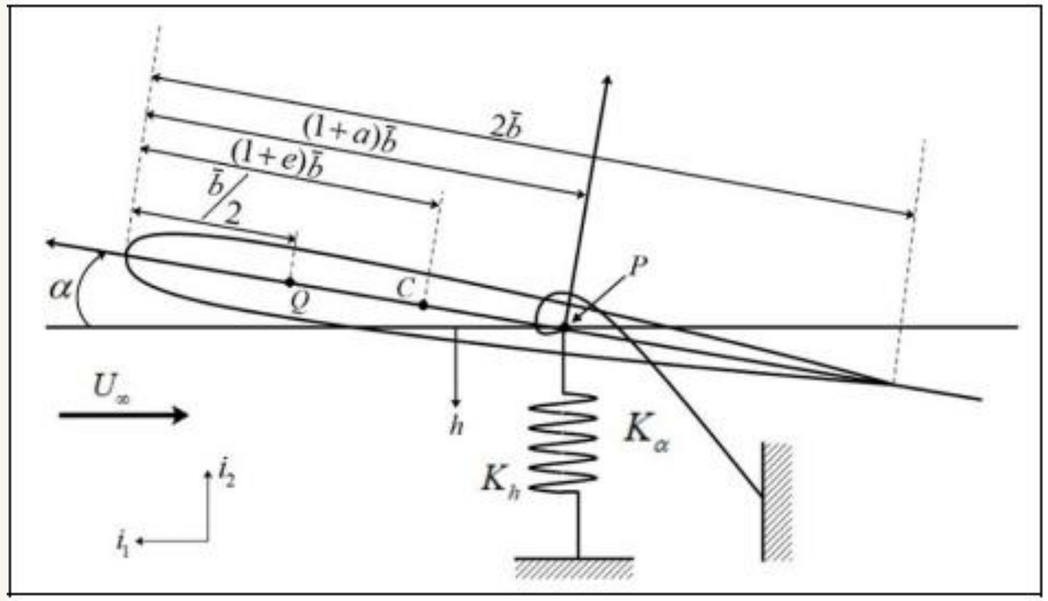


# Methodology



## Load Redistribution

- Typical aeroelastic section – local
- $T(y) = q.c(y)\,(cl(y).ec(y) + cm(y).c(y))$
- Interpolation between aerodynamic and structural mesh - Piecewise Cubic Hermite Interpolating Polynomial
- Geometry update until equilibrium criteria



# Methodology

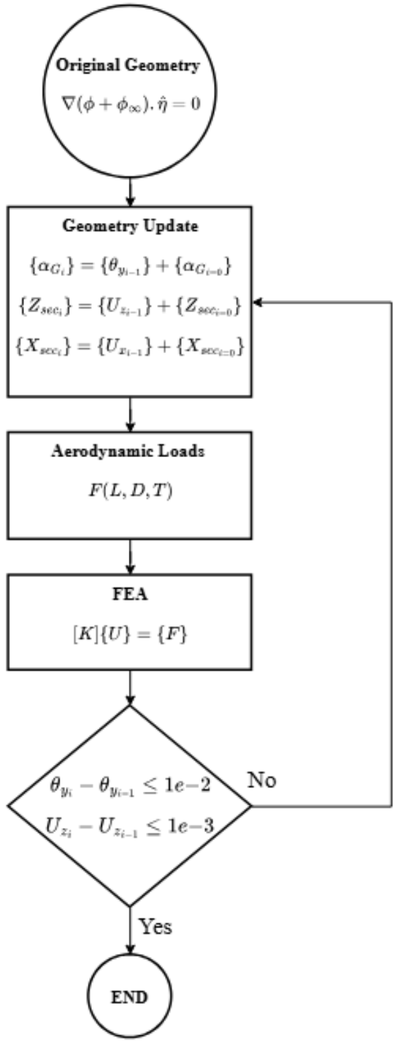


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## Divergence

- Wingtip twist calculated through load redistribution for increasing velocities.

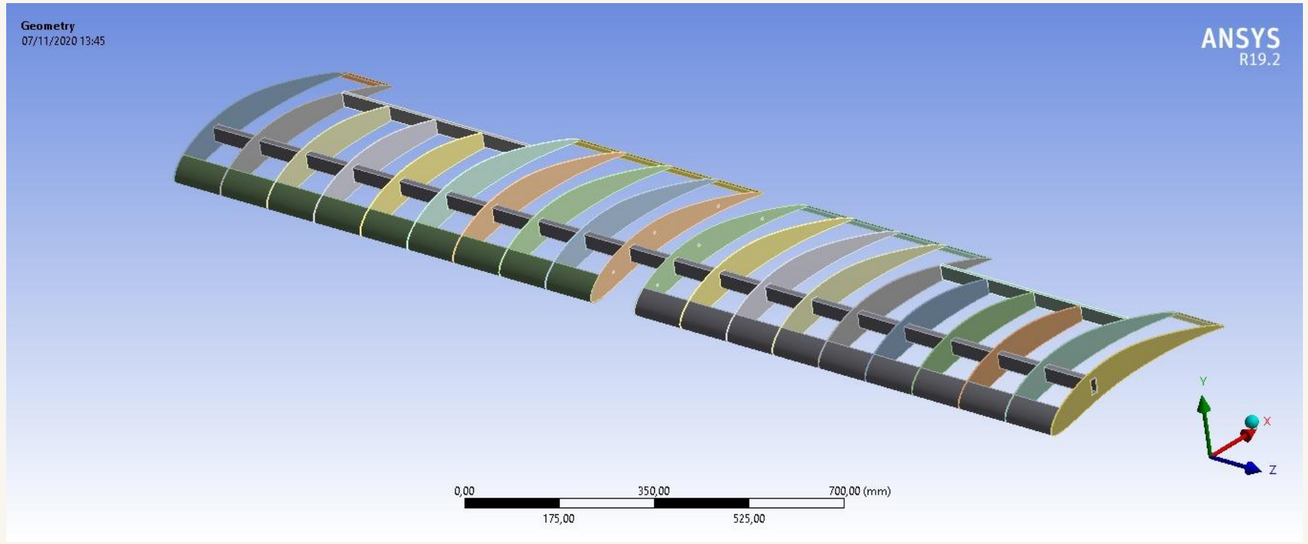


# Methodology



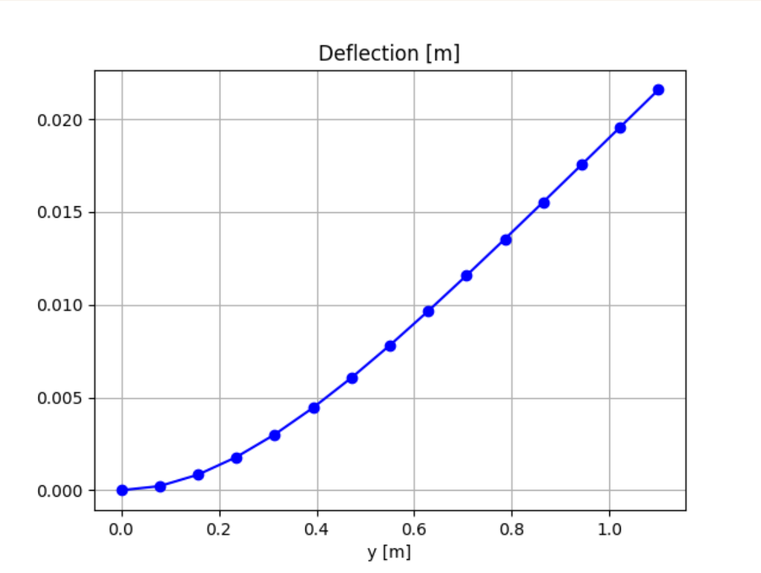
## Study Case

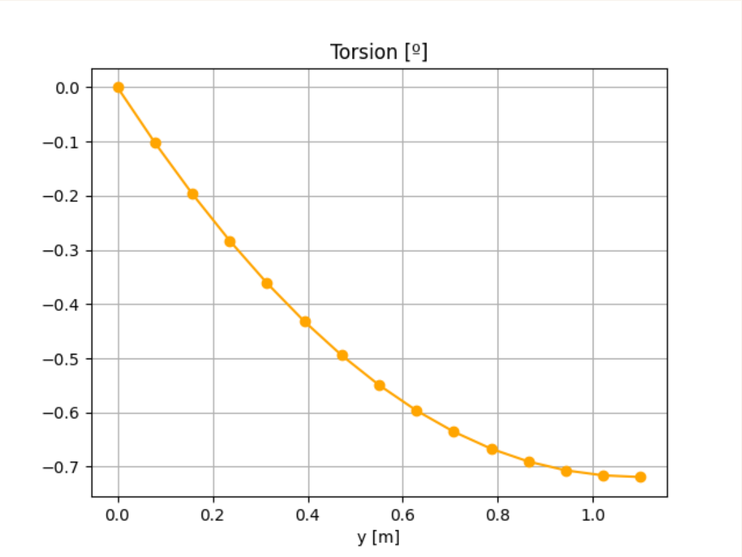
- NI-19 Wing
- Airfoil: E423
- Wingspan: 2.2m
- Chord: 0.49m
- Incidence: 0°
- Reference Results:
  - Deflection: 31.2mm

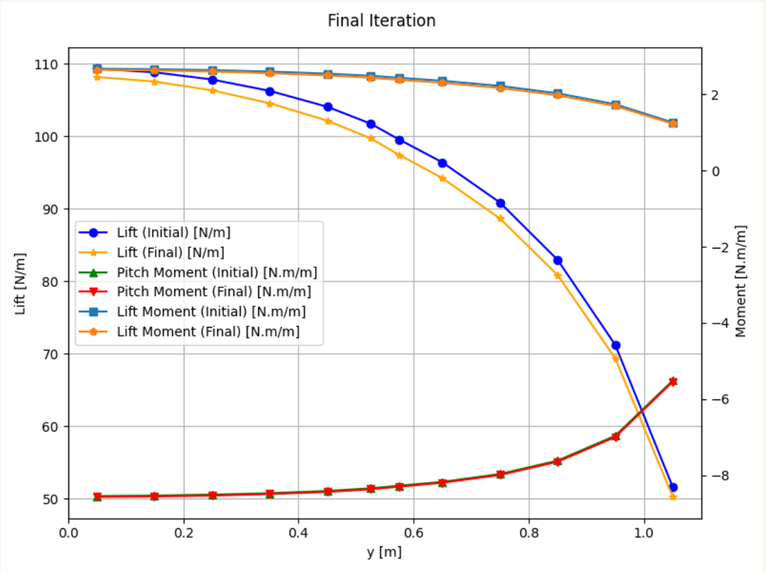


# Results

## Load Redistribution



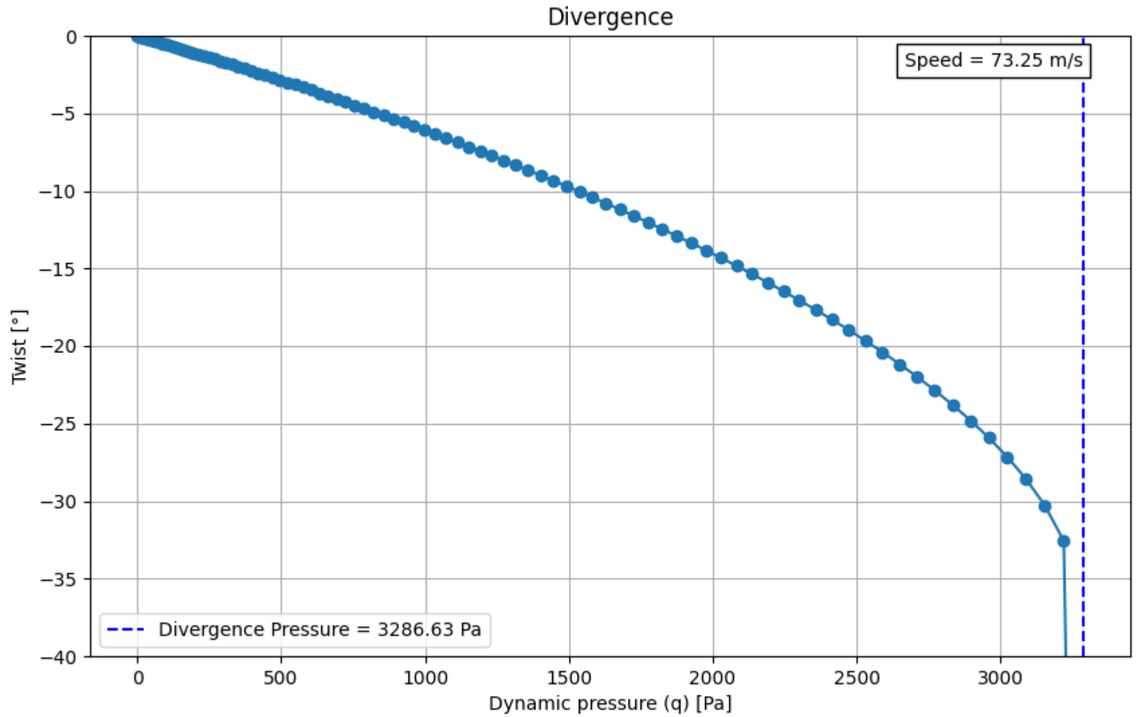




- $C_L$ - 1.87% ↓
- $C_D$ - 4.04% ↓
- $L/D$ - 2.26% ↑

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# Results

## Divergence




- $V_{Div}$ = 73.25 m/s ; Not in the flight envelope

## Remarks

- Deflection lesser than the reference – Elastic load
- Divergence Pressure lesser than theoretical – Low speed, high lift airfoil

# Results

## Remarks

- Deflection lesser than the reference – Elastic load
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